6. $\xi \sim P(x, \theta) = \begin{cases} \frac{\theta-1}{x^\theta}, & x \geq 1 \\ 0, & x < 1 \end{cases}, \quad \theta > 1$

$\vec{x}_n$ – выборка

а) $L(\theta) = P(\vec{x}_n | \theta) = \prod_{i=1}^{n} \frac{\theta-1}{x_i^\theta}$

$\ln L(\theta) = n \ln(\theta-1) - \sum_{i=1}^{n} \theta \ln x_i = n \ln(\theta-1) - \theta \sum_{i=1}^{n} \ln x_i$

$\left(\ln L(\theta-1) - \theta \sum_{i=1}^{n} \ln x_i\right)'_\theta = \frac{n}{\theta-1} - \sum_{i=1}^{n} \ln x_i \Rightarrow \boxed{\theta = 1 + \frac{n}{\sum_{i=1}^{n} \ln x_i}}$

$(\dots)''_{\theta\theta} = -\frac{n}{(\theta-1)^2} < 0$ – максимум

в) Медиана $\equiv$ квантиль $\frac{1}{2}$.

с) $F(x) = \int_1^x \frac{\theta-1}{t^\theta} dt = 1 - x^{-(\theta-1)}$

$F(x_{(\frac{1}{2})}) = \frac{1}{2} \Rightarrow x_{(\frac{1}{2})}^{-(\theta-1)} = \frac{1}{2} \quad x_{(\frac{1}{2})}^{(\theta-1)} = 2 \Rightarrow \boxed{x_{(\frac{1}{2})} = \sqrt[\theta-1]{2}}$

$$x_{(\frac{1}{2})} = 2^{\frac{1}{\theta-1}}$$

Найдём асимпт. довер. интерв. для θ, перейдём к медиане.

$\cancel{I(\theta) = \int_1^\infty \frac{(x^\theta - (\theta-1) x^\theta \ln x)^2}{x^{2\theta}} \frac{\theta-1}{x^\theta} dx =}$

$\cancel{= \int_1^\infty \frac{(1-(\theta-1)\ln x)^2}{x^{3\theta}} (\theta-1) dx =}$

$I(\theta) = \int_1^\infty \left(\frac{\partial}{\partial\theta} \ln P(x,\theta)\right)^2 P(x,\theta) dx = \int_1^\infty \left(\frac{\partial}{\partial\theta}(\ln(\theta-1) - \theta \ln x)\right)^2 \frac{\theta-1}{x^\theta} dx =$

$= \int_1^\infty \left(\frac{1}{\theta-1} - \ln x\right)^2 \frac{\theta-1}{x^\theta} dx = \int_1^\infty \left(\frac{1}{\theta-1} - t\right)^2 \frac{\theta-1}{e^{t\theta}} de^t =$

$\lambda = \theta - 1$

$= \int_1^\infty \left(\frac{1}{\theta-1} - t\right)^2 \frac{(\theta-1)}{e^{(\theta-1)t}} dt = \lambda \int_1^\infty \left(\frac{1}{\lambda^2} - \frac{2t}{\lambda} + t^2\right) e^{-\lambda t} dy$

Берём по частям, получаем:

$$I(\theta) = \frac{1}{\lambda^2} - \frac{2}{\lambda^2} + \frac{2}{\lambda^2} = \frac{1}{\lambda^2}$$

$$\boxed{I(\theta) = \frac{1}{(\theta-1)^2}}$$ - непрер. и больше она 0.

~~Наш~~ $\cancel{\sqrt{n}\,\frac{1+\frac{n}{\sum_{i=1}^{n}\ln x_i} - \theta}{\sqrt{\frac{1}{(\theta-1)^2}}} \sim N(0,1)}$.

$$\cancel{\left(1+\frac{n}{\sum_{i=1}^{n}\ln x_i} - \theta\right)(\theta-1)\sqrt{n} \sim N(0,1)}.$$

$$\cancel{t_{\frac{1-\beta}{2}} < \left(\theta + \frac{n\theta}{\sum_{i=1}^{n}\ln x_i} - \theta^2 - 1 - \frac{n}{\sum \ln x_i} + \theta\right)\sqrt{n} < t_{\frac{1+\beta}{2}}}.$$

$$\cancel{\theta^2 + \left(2 - \frac{n}{\sum_{i=1}^{n}\ln x_i}\right)\theta - \left(1 + \frac{n}{\sum \ln x_i}\right)}$$

$$\int_{-\infty}^{\infty} \frac{d\,\partial F(x,\theta)}{\partial\theta} = \int_1^{\infty} d\, x^{-(\theta-1)} \cdot \ln x = \int_1^{\infty} -\frac{\theta-1}{x^{\theta}}\ln x + \frac{1}{x^{\theta}}\,dx =$$

$$= \int_1^{\infty} \frac{-(\theta-1)\ln x + 1}{x^{\theta}}\,dx = \underbrace{\int_1^{\infty} x^{-\theta}\,dx}_{=\frac{1}{\theta-1}} - (\theta-1)\underbrace{\int_1^{\infty} \ln x\, x^{-\theta}\,dx}_{=\frac{1}{(\theta-1)^2}} = 0$$

$\Rightarrow$ модель регулярна $\Rightarrow$

$$\Rightarrow \sqrt{n}\,\frac{\tilde{\theta} - \theta}{\sigma(\tilde{\theta})} \rightsquigarrow N(0,1)$$

$$\sqrt{n}\,\frac{1+\frac{n}{\sum \ln x_i}-\theta}{\cancel{1}+\frac{n}{\sum \ln x_i}-\cancel{1}} \rightsquigarrow N(0,1)$$

$$t_{\frac{1-\beta}{2}} < \frac{\sum \ln x_i}{\sqrt{n}} + 1 - \theta < t_{\frac{1+\beta}{2}}$$

$$\boxed{1+\frac{\sum \ln x_i}{\sqrt{n}} - t_{\frac{1+\beta}{2}} < \theta < -t_{\frac{1-\beta}{2}} + 1 + \frac{\sum \ln x_i}{\sqrt{n}}}$$ – асимпт. д. инт. ~~оценка~~ θ.

$$\frac{\sum \ln x_i}{\sqrt{n}} - t_{\frac{1+\beta}{2}} < \theta - 1 < \frac{\sum \ln x_i}{\sqrt{n}} - t_{\frac{1-\beta}{2}}$$

$$\frac{1}{\frac{\sum \ln x_i}{\sqrt{n}} - t_{\frac{1-\beta}{2}}} < \frac{1}{\theta - 1} < \frac{1}{\frac{\sum \ln x_i}{\sqrt{n}} - t_{\frac{1+\beta}{2}}}$$

$$\boxed{2\,\frac{1}{\frac{\sum \ln x_i}{\sqrt{n}} - t_{\frac{1-\beta}{2}}} < X_{\left(\frac{1}{2}\right)} < 2\,\frac{1}{\frac{\sum \ln x_i}{\sqrt{n}} - t_{\frac{1+\beta}{2}}}}$$ – асимпт. д. инт. медианы.